

BookStorDB Database Documentation

1. Introduction

The **BookStorDB** database is designed to manage the operations of a book rental store. It tracks information about books, authors, publishers, categories, customers, employees, and rental transactions. The system aims to provide a robust and organized structure for managing inventory, user accounts, and rental activities.

2. Database Schema Overview

The **BookStorDB** comprises eight main tables, each serving a specific function in the overall system:

- **Authors:** Stores information about book authors.
- **Books:** Contains details about the books available in the store.
- **Categories:** Organizes books into different genres or subjects.
- **Customers:** Manages customer accounts and their details.
- **Employee:** Stores information about the bookstore employees.
- **Publishers:** Holds details about the publishers of the books.
- **Rentals:** Records rental transactions between customers and books.
- **Role:** Defines different user roles within the system (e.g., Author, Employee, Customer).

3. Table Details

3.1. **Role** Table

This table defines the different roles that users (Authors, Employees, Customers) can have within the system.

Column Name	Data Type	Nullable	Description
Role_ID	int	NO	Primary key, unique identifier for the role. Auto-incrementing starting from 400.
Role_Name	nvarchar(30)	NO	Name of the role (e.g., 'Author', 'Employee', 'Customer').
Date_Given	date	YES	Date when the role was assigned.

Primary Key: Role_ID

3.2. Authors Table

Stores information about the authors of the books.

Column Name	Data Type	Nullable	Description
Author_Id	int	NO	Primary key, unique identifier for the author. Auto-incrementing starting from 100.
Author_Name	nvarchar(70)	NO	Full name of the author.
Author_Email	varchar(50)	NO	Email address of the author.
Role_ID	int	YES	Foreign key referencing Role_ID in the Role table. Indicates the author's role.
Author_Password	nvarchar(70)	YES	Password for the author's account.

Primary Key: Author_Id **Foreign Key:** Role_ID references Role(Role_ID)

3.3. Publishers Table

Contains details about the publishers of the books.

Column Name	Data Type	Nullable	Description
Publisher_Id	int	NO	Primary key, unique identifier for the publisher. Auto-incrementing.
Publisher_Name	nvarchar(150)	NO	Name of the publisher.
Publisher_Contact	nvarchar(70)	YES	Contact person or phone number for the publisher.
Publisher_Address	nvarchar(100)	YES	Address of the publisher.

Primary Key: Publisher_Id

3.4. Categories Table

Categorizes books into different genres or subjects.

Column Name	Data Type	Nullable	Description
Category_Id	int	NO	Primary key, unique identifier for the category. Auto-incrementing.
Category_Name	nvarchar(100)	NO	Name of the category (e.g., 'Science', 'Technology').
Description	nvarchar(150)	YES	Brief description of the category.

Primary Key: Category_Id

3.5. Books Table

Stores detailed information about each book available in the store.

Column Name	Data Type	Nullable	Description
Book_Id	int	NO	Primary key, unique identifier for the book. Auto-incrementing.
Book_Title	nvarchar(200)	NO	Title of the book.
Book_ISBN	nvarchar(50)	YES	International Standard Book Number, unique identifier for the book.
CopiesAvailable	int	YES	Number of copies of the book currently available for rent. Default is 1.
Author_Id	int	YES	Foreign key referencing Author_Id in the Authors table.
Publisher_Id	int	YES	Foreign key referencing Publisher_Id in the Publishers table.
Category_Id	int	YES	Foreign key referencing Category_Id in the Categories table.

Primary Key: Book_Id **Unique Constraint:** Book_ISBN **Foreign Keys:**

- Author_Id references Authors(Author_Id)
- Publisher_Id references Publishers(Publisher_Id)
- Category_Id references Categories(Category_Id)

3.6. customers Table

Manages customer accounts and their personal details.

Column Name	Data Type	Nullable	Description
Customer_ID	int	NO	Primary key, unique identifier for the customer. Auto-incrementing starting from 500.
Customer_Name	nvarchar(70)	NO	Full name of the customer.
Customer_Email	varchar(60)	NO	Email address of the customer.
Role_ID	int	YES	Foreign key referencing Role_ID in the Role table. Indicates the customer's role.
Customer_Password	nvarchar(100)	YES	Password for the customer's account.

Primary Key: Customer_ID **Foreign Key:** Role_ID references Role(Role_ID)

3.7. Employee Table

Stores information about the employees working at the bookstore.

Column Name	Data Type	Nullable	Description
Employee_Id	int	NO	Primary key, unique identifier for the employee. Auto-incrementing starting from 200.
Employee_FullName	nvarchar(100)	NO	Full name of the employee.
Employee_Email	nvarchar(100)	NO	Email address of the employee.
Employee_Address	nvarchar(100)	YES	Residential address of the employee.
Employee_Gender	bit	YES	Gender of the employee (e.g., 0 for Female, 1 for Male).
Employee_Phone	nvarchar(20)	NO	Phone number of the employee.
Employee_JoinDate	datetime	YES	Date when the employee joined the company.
Role_ID	int	YES	Foreign key referencing Role_ID in the Role table. Indicates the employee's role.
Employee_Password	nvarchar(100)	YES	Password for the employee's account.

Primary Key: Employee_Id **Foreign Key:** Role_ID references Role(Role_ID)

3.8. Rentals Table

Records details of book rental transactions.

Column Name	Data Type	Nullable	Description
Rental_Id	int	NO	Primary key, unique identifier for the rental transaction. Auto-incrementing starting from 300.
Customer_ID	int	YES	Foreign key referencing Customer_ID in the Customers table.
Book_Id	int	YES	Foreign key referencing Book_Id in the Books table.
Rent_Date	datetime	NO	Date and time when the book was rented. Default is current date/time.
Return_Date	datetime	YES	Date and time when the book is expected or was returned.

Primary Key: Rental_Id **Foreign Keys:**

- Customer_ID references Customers(Customer_ID)
- Book_Id references Books(Book_Id)

4. System Logic and Relationships

The BookStorDB is designed with clear relationships to ensure data integrity and efficient querying:

- Roles and Users:** The Role table acts as a central authority for defining user types. Authors, Customers, and Employee tables all link to the Role table via Role_ID, allowing for flexible role management and access control.
- Books and Metadata:** Books are central to the system. Each book is associated with an Author, a Publisher, and a Category through foreign key relationships. This structure allows for detailed classification and retrieval of book information.
- Rental Process:** The Rentals table connects Customers with Books. When a book is rented, a record is created in Rentals, linking a specific Customer_ID to a Book_Id with Rent_Date and an expected Return_Date. The

`CopiesAvailable` in the `Books` table is decremented upon rental and would typically be incremented upon return (though the return logic is handled by stored procedures, not directly in the schema).

- **Identity Management:** All primary keys are auto-incrementing, simplifying data entry and ensuring uniqueness. Passwords are stored for Authors, Customers, and Employees, indicating a login mechanism for different user types.

5. Entity-Relationship Diagram (ERD)

Below is the Entity-Relationship Diagram for the `BookStorDB` database, illustrating the tables and their relationships.